1 Implementing Naive Bayes Theorem

[66]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report
```

[67]:
```
# Load the dataset
df = pd.read_excel('Naive_Bayes_Dataset.xlsx')  # Adjust path as needed
```

[68]:
```
df
```

[68]:

	Day	Discount	Free_Delivery	Purchase
0	Weekday	Yes	Yes	Yes
1	Weekday	Yes	Yes	Yes
2	Weekday	No	No	No
3	Holiday	Yes	Yes	Yes
4	Weekend	Yes	Yes	Yes
5	Holiday	No	No	No
6	Weekend	Yes	No	Yes
7	Weekday	Yes	Yes	Yes
8	Weekend	Yes	Yes	Yes
9	Holiday	Yes	Yes	Yes
10	Holiday	No	Yes	Yes
11	Holiday	No	No	No
12	Weekend	Yes	Yes	Yes
13	Holiday	Yes	Yes	Yes
14	Holiday	Yes	Yes	Yes
15	Weekday	Yes	Yes	Yes
16	Holiday	No	Yes	Yes
17	Weekday	Yes	No	Yes
18	Weekend	No	No	Yes
19	Weekend	No	Yes	Yes
20	Weekday	Yes	Yes	Yes
21	Weekend	Yes	Yes	No
22	Holiday	No	Yes	Yes

```
23  Weekday      Yes           Yes      Yes
24  Holiday       No            No       No
25  Weekday       No           Yes       No
26  Weekday      Yes           Yes      Yes
27  Weekday      Yes           Yes      Yes
28  Holiday      Yes           Yes      Yes
29  Weekend      Yes           Yes      Yes
```

[69]:
```
# Encode categorical variables
label_encoders = {}
for column in ['Day', 'Discount', 'Free_Delivery', 'Purchase']:
    le = LabelEncoder()
    df[column] = le.fit_transform(df[column])
    label_encoders[column] = le  # Save encoder for future decoding if needed
```

[70]:
```
# Split data into features (X) and target (y)
X = df[['Day', 'Discount', 'Free_Delivery']]
y = df['Purchase']
```

[71]:
```
# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42)
```

[72]:
```
# Initialize and train the Naive Bayes model
nb_model = CategoricalNB()
nb_model.fit(X_train, y_train)
```

[72]:
```
CategoricalNB()
```

[73]:
```
# Make predictions on the test set
y_pred = nb_model.predict(X_test)
```

[74]:
```
# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
              precision    recall  f1-score   support

           1       1.00      1.00      1.00         6

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6
```

[75]:
```
# Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['No']})
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
prediction = nb_model.predict(example)
print("Purchase Prediction:", label_encoders['Purchase'].
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

[76]:
```
# Check if "Holiday" exists in the Day encoder classes
print("Classes for Day:", label_encoders['Day'].classes_)
```

```
Classes for Day: ['Holiday' 'Weekday' 'Weekend']
```

[77]:
```
# Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['No']})
# This line creates a DataFrame named example with a single row representing an␣
↪example input.

# Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and␣
↪transforms the categorical
#text values into numeric codes, which the model requires as input.


# Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a prediction
# on the transformed example input.


# Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

[79]:
```
# Example prediction
example = pd.DataFrame({'Day': ['Weekend'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['Yes']})
# This line creates a DataFrame named example with a single row representing an␣
↪example input.
```

```python
# Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and␣
↪transforms the categorical
#text values into numeric codes, which the model requires as input.


# Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a prediction
# on the transformed example input.


# Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

2.0 Naive Bayes Classifier -Titanic

In [1]:
```
import pandas as pd
```

In [2]:
```
df= pd.read_csv('titanic.csv')
df.head()
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

```
# removing null values of Age// better if we remove later on the updated dataset
```

In [4]:
```
handle = df['Age'].mean()
```

In [5]:
```
handle
```

Out[5]:
```
30.272590361445783
```

In [6]:
```
df.Age=  df.Age.fillna(handle)
```

In [7]:
```
df
```

Out[7]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	0	3	Kelly, Mr. James	male	34.50000	0	0	330911	7.8292	NaN	Q
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.00000	1	0	363272	7.0000	NaN	S
2	894	0	2	Myles, Mr. Thomas Francis	male	62.00000	0	0	240276	9.6875	NaN	Q
3	895	0	3	Wirz, Mr. Albert	male	27.00000	0	0	315154	8.6625	NaN	S
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.00000	1	1	3101298	12.2875	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
413	1305	0	3	Spector, Mr. Woolf	male	30.27259	0	0	A.5. 3236	8.0500	NaN	S
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	39.00000	0	0	PC 17758	108.9000	C105	C
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	38.50000	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	1308	0	3	Ware, Mr. Frederick	male	30.27259	0	0	359309	8.0500	NaN	S
417	1309	0	3	Peter, Master. Michael J	male	30.27259	1	1	2668	22.3583	NaN	C

418 rows × 12 columns

Assumption: Make a naive assumption that features such as male,class,age,cabin,fare, etc. are independant of each other

In [8]:
```
df.drop(['PassengerId','Name','SibSp','Parch','Ticket','Cabin','Embarked'],axis ='columns',inplace= True
```

In [9]:
```
df.head()
```

Out[9]:

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	34.5	7.8292
1	1	3	female	47.0	7.0000
2	0	2	male	62.0	9.6875
3	0	3	male	27.0	8.6625
4	1	3	female	22.0	12.2875

1 - survived 0 - not survived , At the end we will work on final obtained from inputs and targer for train_test_split()

In [16]:
```
# 1 - survived 0 - not survived
inputs= df.drop('Survived', axis = 'columns')
target = df.Survived
```

Encoding Sex

In [10]:
```
dummies = pd.get_dummies(df['Sex'])
```

In [12]:
```
dummies = dummies.astype(int)
# will display o and 1 instead of True and False
```

In [13]:
```
dummies
```

Out[13]:

	female	male
0	0	1
1	1	0
2	0	1
3	0	1
4	1	0
...	...	...
413	0	1
414	1	0
415	0	1
416	0	1
417	0	1

418 rows × 2 columns

In [ ]:
```
#inputs = inputs.drop('Sex','male' axis='columns')
```

In [14]:
```
df
```

Out[14]:

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	34.50000	7.8292
1	1	3	female	47.00000	7.0000
2	0	2	male	62.00000	9.6875
3	0	3	male	27.00000	8.6625
4	1	3	female	22.00000	12.2875
...	...	...	...	...	...
413	0	3	male	30.27259	8.0500
414	1	1	female	39.00000	108.9000
415	0	3	male	38.50000	7.2500
416	0	3	male	30.27259	8.0500
417	0	3	male	30.27259	22.3583

418 rows × 5 columns

In [17]:
```
inputs
```

Out[17]:

	Pclass	Sex	Age	Fare
0	3	male	34.50000	7.8292
1	3	female	47.00000	7.0000
2	2	male	62.00000	9.6875
3	3	male	27.00000	8.6625
4	3	female	22.00000	12.2875
...	...	...	...	...
413	3	male	30.27259	8.0500
414	1	female	39.00000	108.9000
415	3	male	38.50000	7.2500
416	3	male	30.27259	8.0500
417	3	male	30.27259	22.3583

418 rows × 4 columns

In [21]:
```
inputs = inputs.drop(['Sex'] ,axis='columns')
```

In [22]:
```
inputs
```

Out[22]:

	Pclass	Age	Fare
0	3	34.50000	7.8292
1	3	47.00000	7.0000
2	2	62.00000	9.6875
3	3	27.00000	8.6625
4	3	22.00000	12.2875
...	...	...	...
413	3	30.27259	8.0500
414	1	39.00000	108.9000
415	3	38.50000	7.2500
416	3	30.27259	8.0500
417	3	30.27259	22.3583

418 rows × 3 columns

In [23]:
```
#concatening columns with sex column
```

In [24]:
```
df
```

Out[24]:

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	34.50000	7.8292
1	1	3	female	47.00000	7.0000
2	0	2	male	62.00000	9.6875
3	0	3	male	27.00000	8.6625
4	1	3	female	22.00000	12.2875
...	...	...	...	...	...
413	0	3	male	30.27259	8.0500
414	1	1	female	39.00000	108.9000
415	0	3	male	38.50000	7.2500
416	0	3	male	30.27259	8.0500
417	0	3	male	30.27259	22.3583

418 rows × 5 columns

In [25]:
```
inputs
```

Out[25]:

	Pclass	Age	Fare
0	3	34.50000	7.8292
1	3	47.00000	7.0000
2	2	62.00000	9.6875
3	3	27.00000	8.6625
4	3	22.00000	12.2875
...	...	...	...
413	3	30.27259	8.0500
414	1	39.00000	108.9000
415	3	38.50000	7.2500
416	3	30.27259	8.0500
417	3	30.27259	22.3583

418 rows × 3 columns

In [26]:
```
merged = pd.concat([inputs,dummies],axis = 'columns')
```

In [27]:
```
merged
```

Out[27]:

	Pclass	Age	Fare	female	male
0	3	34.50000	7.8292	0	1
1	3	47.00000	7.0000	1	0
2	2	62.00000	9.6875	0	1
3	3	27.00000	8.6625	0	1
4	3	22.00000	12.2875	1	0
...	...	...	...	...	...
413	3	30.27259	8.0500	0	1
414	1	39.00000	108.9000	1	0
415	3	38.50000	7.2500	0	1
416	3	30.27259	8.0500	0	1
417	3	30.27259	22.3583	0	1

418 rows × 5 columns

In [30]:
```
final = merged.drop(['male'],axis ='columns')
```

In [31]:
```
final
```

Out[31]:

	Pclass	Age	Fare	female
0	3	34.50000	7.8292	0
1	3	47.00000	7.0000	1
2	2	62.00000	9.6875	0
3	3	27.00000	8.6625	0
4	3	22.00000	12.2875	1
...	...	...	...	...
413	3	30.27259	8.0500	0
414	1	39.00000	108.9000	1
415	3	38.50000	7.2500	0
416	3	30.27259	8.0500	0
417	3	30.27259	22.3583	0

418 rows × 4 columns

In [34]:
```
final.columns[final.isna().any()]
```

Out[34]:
```
Index(['Fare'], dtype='object')
```

In [39]:
```
null_count_city = df['Fare'].isnull().sum()
print(f"Number of null values in 'Fare' column: {null_count_city}")
```

```
Number of null values in 'Fare' column: 1
```

In [38]:
```
final.Fare[:50]
```

Out[38]:
```
0       7.8292
1       7.0000
2       9.6875
3       8.6625
4      12.2875
5       9.2250
6       7.6292
7      29.0000
8       7.2292
9      24.1500
10      7.8958
11     26.0000
12     82.2667
13     26.0000
14     61.1750
15     27.7208
16     12.3500
17      7.2250
18      7.9250
19      7.2250
20     59.4000
21      3.1708
22     31.6833
23     61.3792
24    262.3750
25     14.5000
26     61.9792
27      7.2250
28     30.5000
29     21.6792
30     26.0000
31     31.5000
32     20.5750
33     23.4500
34     57.7500
35      7.2292
36      8.0500
37      8.6625
38      9.5000
39     56.4958
40     13.4167
41     26.5500
42      7.8500
43     13.0000
44     52.5542
45      7.9250
46     29.7000
47      7.7500
48     76.2917
49     15.9000
Name: Fare, dtype: float64
```

In [40]:
```
final.Fare = final.Fare.fillna(final.Fare.mean())
final.head()
```

Out[40]:

	Pclass	Age	Fare	female
0	3	34.5	7.8292	0
1	3	47.0	7.0000	1
2	2	62.0	9.6875	0
3	3	27.0	8.6625	0
4	3	22.0	12.2875	1

In [41]:
```
from sklearn.model_selection import train_test_split
```

In [43]:
```
xtrain, xtest, ytrain, ytest = train_test_split(final, target, test_size = 0.30, random_state =1)
```

In [44]:
```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB ()
```

In [47]:
```
model.fit(xtrain, ytrain)
```

Out[47]:
```
▾ GaussianNB
GaussianNB()
```

In [49]:
```
model.score(xtest,ytest)
```

Out[49]:
```
1.0
```

In [50]:
```
xtest[:10]
```

Out[50]:

	Pclass	Age	Fare	female
358	3	30.27259	7.7500	0
164	2	41.00000	13.0000	0
17	3	21.00000	7.2250	0
67	1	47.00000	42.4000	0
4	3	22.00000	12.2875	1
377	2	21.00000	11.5000	0
214	3	38.00000	7.7750	1
290	1	30.27259	39.6000	0
381	3	26.00000	7.8792	0
5	3	14.00000	9.2250	0

In [53]:
```
ytest[0:10]
```

Out[53]:
```
358    0
164    0
17     0
67     0
4      1
377    0
214    1
290    0
381    0
5      0
Name: Survived, dtype: int64
```

In [54]:
```
model.predict(xtest[0:10])
```

Out[54]:
```
array([0, 0, 0, 0, 1, 0, 1, 0, 0, 0], dtype=int64)
```

In [55]:
```
model.predict_proba(xtest[:10])
```

Out[55]:
```
array([[1., 0.],
       [1., 0.],
       [1., 0.],
       [1., 0.],
       [0., 1.],
       [1., 0.],
       [0., 1.],
       [1., 0.],
       [1., 0.],
       [1., 0.]])
```

In [62]:
```
#pclass, Age, Fare, Gender

test = [[1,23.000000,113.2750,0]]
#test = xtest
a= model.predict(test)
if a[0] == 0:
      print("Not Survived")
else:
       print(" Survived")
```

```
Not Survived

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning: X does not have valid feat
ure names, but GaussianNB was fitted with feature names
  warnings.warn(
```

In [ ]:

Calculate value with cross validation

In [63]:
```
from sklearn.model_selection import cross_val_score
```

In [65]:
```
cross_val_score (GaussianNB(),xtrain,ytrain,cv = 5)
```

Out[65]:
```
array([1., 1., 1., 1., 1.])
```

In [ ]:

feature-selection-stdm

August 12, 2025

1 Univariate Selecetion Method @Feature Selection, Algorithm: SelectKBest Algorithm

[22]:
```
import pandas as pd
import numpy as np
import seaborn as sns
```

[16]:
```
#df = pd.read_csv("cardio_train.csv",sep = ';')
#df = pd.read_csv("bank.csv",sep = ';')

df = pd.read_csv("cardio_train.csv",sep = ';')
```

[17]:
```
df
```

[17]:
```
          id    age  gender  height  weight  ap_hi  ap_lo  cholesterol  gluc  \
0          0  18393       2     168    62.0    110     80            1     1
1          1  20228       1     156    85.0    140     90            3     1
2          2  18857       1     165    64.0    130     70            3     1
3          3  17623       2     169    82.0    150    100            1     1
4          4  17474       1     156    56.0    100     60            1     1
…        …     …       …       …       …      …      …            …     …
69995  99993  19240       2     168    76.0    120     80            1     1
69996  99995  22601       1     158   126.0    140     90            2     2
69997  99996  19066       2     183   105.0    180     90            3     1
69998  99998  22431       1     163    72.0    135     80            1     2
69999  99999  20540       1     170    72.0    120     80            2     1

       smoke  alco  active  cardio
0          0     0       1       0
1          0     0       1       1
2          0     0       0       1
3          0     0       1       1
4          0     0       0       0
…        …   …       …     …
69995      1     0       1       0
69996      0     0       1       1
69997      0     1       0       1
```

```
69998      0     0       0       1
69999      0     0       1       0

[70000 rows x 13 columns]
```

[19]:
```
df.shape
```

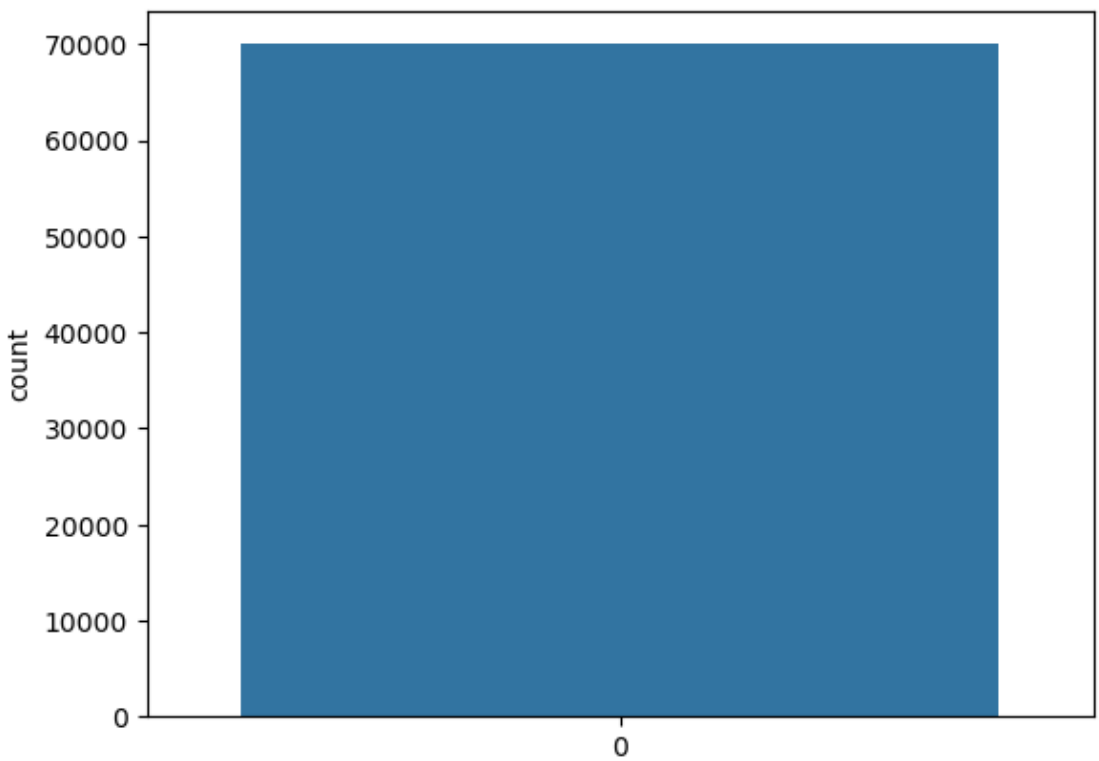
[19]:
```
(70000, 13)
```

[20]:
```
df["cardio"].value_counts()
```

[20]:
```
cardio
0    35021
1    34979
Name: count, dtype: int64
```

[24]:
```
sns.countplot(df["cardio"])
```

[24]:
```
<Axes: ylabel='count'>
```



[25]:
```
#          x = df.iloc[:, :-1]:
#This line is selecting all rows (:) and all columns except the last one (:-1).
```

```
# : means "select all rows."
# :-1 means "select all columns except the last one." In Python slicing, :-1␣
↪excludes the last item, so this will include all columns up to, but not␣
↪including, the last one.
# Result: x will be a DataFrame containing all the columns of df except the␣
↪last column.


x = df.iloc[: , :-1]
```

[27]:
```
# y = df.iloc[:, 12]
# This line is selecting all rows (:) and the 13th column (remember, indexing␣
↪in Python starts at 0, so index 12 corresponds to the 13th column).
#  : means "select all rows."
# 12 means "select the 13th column" (as Python uses 0-based indexing).
# Result: y will be a Series containing the values of the 13th column of df.
```

```
  Cell In[27], line 7
    y = df.iloc[:, 12]
    ^
IndentationError: unexpected indent
```

[28]:
```
#y = df.iloc[:, 12]
y = df.iloc [:,12]
```

[29]:
```
from sklearn.feature_selection import SelectKBest
```

[30]:
```
from sklearn.feature_selection import f_classif
```

[31]:
```
FIT_FEATURES= SelectKBest(score_func = f_classif)
```

[32]:
```
FIT_FEATURES.fit(x,y)
```

[32]:
```
SelectKBest()
```

[34]:
```
pd.DataFrame(FIT_FEATURES.scores_)
```

[34]:

	0
0	1.010461
1	4209.007957
2	4.603641
3	8.197397
4	2388.777887
5	208.339524
6	303.629011

```
7   3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

[35]:
```
score_COL = pd.DataFrame(FIT_FEATURES.scores_)
```

[36]:
```
score_COL
```

[36]:
```
              0
0      1.010461
1   4209.007957
2      4.603641
3      8.197397
4   2388.777887
5    208.339524
6    303.629011
7   3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

[42]:
```
# Assuming FIT_FEATURES_scores is an array or a list of scores
score_COL = pd.DataFrame(FIT_FEATURES.scores_, columns=["scorevalue"])
# Now score_COL is a DataFrame with a single column named "scorevalue"
```

[43]:
```
score_COL
```

[43]:
```
     scorevalue
0      1.010461
1   4209.007957
2      4.603641
3      8.197397
4   2388.777887
5    208.339524
6    303.629011
7   3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

[46]:
```
NEW_COL = pd.DataFrame(x.columns)
```

[47]:
```
NEW_COL
```

[47]:

```
                0
0              id
1             age
2          gender
3          height
4          weight
5           ap_hi
6           ap_lo
7    cholesterol
8            gluc
9           smoke
10           alco
11         active
```

[52]:

```
top_features = pd.concat([NEW_COL, score_COL], axis=1)
```

[53]:

```
top_features
```

[53]:

```
                0   scorevalue
0              id     1.010461
1             age  4209.007957
2          gender     4.603641
3          height     8.197397
4          weight  2388.777887
5           ap_hi   208.339524
6           ap_lo   303.629011
7    cholesterol  3599.361137
8            gluc   562.772977
9           smoke    16.790541
10           alco     3.761355
11         active    89.091494
```

[55]:

```
top_features.nlargest (8,"scorevalue")
```

[55]:

```
                0   scorevalue
1             age  4209.007957
7    cholesterol  3599.361137
4          weight  2388.777887
8            gluc   562.772977
6           ap_lo   303.629011
5           ap_hi   208.339524
11         active    89.091494
9           smoke    16.790541
```

eature-selection-using-correlation

August 12, 2025

1 Filter Method

[18]:
```
import pandas as pd

# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

[19]:
```
df
```

[19]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[20]:
```
# Check for non-numeric data in the columns
print(df.dtypes)
```

```
location       object
area           object
bedroom         int64
bathroom        int64
owner          object
blodd group    object
price           int64
dtype: object
```

[21]:
```
# Identify non-numeric values in specific columns
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

```
Non-numeric values in 'location ':
 ['a' 'b' 'c' 'd' 'e' 'f' 'g' 'h' 'i' 'j']
Non-numeric values in 'area ':
 ['dhaka' 'khulna' 'rajshahi' 'cumilla' 'barishal' 'natore']
Non-numeric values in 'owner ':
 ['x' 'y' 'p' 'q' 'w' 'l' 'r' 's' 'k' 'v']
Non-numeric values in 'blodd group':
 ['o+' 'ab+' 'ab-']
```

[22]:
```python
# Option 1: Convert columns to numeric, forcing errors to NaN
df = df.apply(pd.to_numeric, errors='coerce')
```

[23]:
```python
df
```

[23]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	NaN	NaN	3	2	NaN	NaN	10000
1	NaN	NaN	4	2	NaN	NaN	12000
2	NaN	NaN	2	2	NaN	NaN	11000
3	NaN	NaN	3	2	NaN	NaN	15000
4	NaN	NaN	3	1	NaN	NaN	12000
5	NaN	NaN	2	1	NaN	NaN	11000
6	NaN	NaN	2	1	NaN	NaN	10000
7	NaN	NaN	2	3	NaN	NaN	15000
8	NaN	NaN	3	3	NaN	NaN	12000
9	NaN	NaN	4	2	NaN	NaN	11000

[ ]:
```python
# Option 2: Alternatively, remove or replace non-numeric values
# Replace non-numeric values with NaN
df.replace(['a', 'N/A', 'unknown'], float('nan'), inplace=True)

# Drop rows with NaN values (if appropriate)
df.dropna(inplace=True)
```

[10]:
```python
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[10], line 2
      1 # Calculate correlation with the target variable 'price'
----> 2 correlation = df.corr()

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:10054, in␣
 ↪DataFrame.corr(self, method, min_periods, numeric_only)
  10052 cols = data.columns
  10053 idx = cols.copy()
> 10054 mat = data.to_numpy(dtype=float, na_value=np.nan, copy=False)
```

```
  10056 if method == "pearson":
  10057     correl = libalgos.nancorr(mat, minp=min_periods)

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:1838, in␣
 ↪DataFrame.to_numpy(self, dtype, copy, na_value)
   1836 if dtype is not None:
   1837     dtype = np.dtype(dtype)
-> 1838 result = self._mgr.as_array(dtype=dtype, copy=copy, na_value=na_value)
   1839 if result.dtype is not dtype:
   1840     result = np.array(result, dtype=dtype, copy=False)

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\internals\managers.
 ↪py:1732, in BlockManager.as_array(self, dtype, copy, na_value)
   1730         arr.flags.writeable = False
   1731 else:
-> 1732     arr = self._interleave(dtype=dtype, na_value=na_value)
   1733     # The underlying data was copied within _interleave, so no need
   1734     # to further copy if copy=True or setting na_value
   1736 if na_value is not lib.no_default:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\internals\managers.
 ↪py:1794, in BlockManager._interleave(self, dtype, na_value)
   1792     else:
   1793         arr = blk.get_values(dtype)
-> 1794     result[rl.indexer] = arr
   1795     itemmask[rl.indexer] = 1
   1797 if not itemmask.all():

ValueError: could not convert string to float: 'a'
```

[ ]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

[6]:
```
print(df.dtypes)
```

```
location       object
area           object
bedroom         int64
bathroom        int64
owner          object
blodd group    object
dtype: object
```

[14]:
```
# Option 1: Convert columns to numeric, forcing errors to NaN
df = df.apply(pd.to_numeric, errors='coerce')
```

[15]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

[16]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
 price          1.000000
bathroom       0.495798
bedroom       -0.015721
location            NaN
area                NaN
owner               NaN
blodd group         NaN
Name: price, dtype: float64
```

[17]:
```
df
```

[17]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	NaN	NaN	3	2	NaN	NaN	10000
1	NaN	NaN	4	2	NaN	NaN	12000
2	NaN	NaN	2	2	NaN	NaN	11000
3	NaN	NaN	3	2	NaN	NaN	15000
4	NaN	NaN	3	1	NaN	NaN	12000
5	NaN	NaN	2	1	NaN	NaN	11000
6	NaN	NaN	2	1	NaN	NaN	10000
7	NaN	NaN	2	3	NaN	NaN	15000
8	NaN	NaN	3	3	NaN	NaN	12000
9	NaN	NaN	4	2	NaN	NaN	11000

[24]:
```
from sklearn.preprocessing import OrdinalEncoder
```

[26]:
```
categorical_column = 'location'
```

[27]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[27], line 2
      1 encoder = OrdinalEncoder()
----> 2 df[categorical_column] = encoder.fit_transform(df[[categorical_column]]

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:3767, in␣
 ↪DataFrame.__getitem__(self, key)
   3765     if is_iterator(key):
```

```
   3766         key = list(key)
-> 3767     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   3769 # take() does not accept boolean indexers
   3770 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
 ↪5877, in Index._get_indexer_strict(self, key, axis_name)
   5874 else:
   5875     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 5877 self._raise_if_missing(keyarr, indexer, axis_name)
   5879 keyarr = self.take(indexer)
   5880 if isinstance(key, Index):
   5881     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
 ↪5938, in Index._raise_if_missing(self, key, indexer, axis_name)
   5936     if use_interval_msg:
   5937         key = list(key)
-> 5938     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   5940 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
   5941 raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['location'], dtype='object')] are in the [columns]"
```

2 Start from here

[28]:
```
import pandas as pd

# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

[29]:
```
df
```

[29]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[30]:
```
# Check for non-numeric data in the columns
print(df.dtypes)
```

```
location       object
area           object
bedroom         int64
bathroom        int64
owner          object
blodd group    object
price           int64
dtype: object
```

[31]:
```
# Identify non-numeric values in specific columns
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

```
Non-numeric values in 'location ':
 ['a' 'b' 'c' 'd' 'e' 'f' 'g' 'h' 'i' 'j']
Non-numeric values in 'area ':
 ['dhaka' 'khulna' 'rajshahi' 'cumilla' 'barishal' 'natore']
Non-numeric values in 'owner ':
 ['x' 'y' 'p' 'q' 'w' 'l' 'r' 's' 'k' 'v']
Non-numeric values in 'blodd group':
 ['o+' 'ab+' 'ab-']
```

3 Encoding / Handling another column => location

[32]:
```
from sklearn.preprocessing import OrdinalEncoder
```

[36]:
```
categorical_column = 'location'
```

[35]:
```
df
```

[35]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[34]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[34], line 2
      1 encoder = OrdinalEncoder()
----> 2 df[categorical_column] = encoder.fit_transform(df[[categorical_column]]

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:3767, in␣
 ↪DataFrame.__getitem__(self, key)
   3765     if is_iterator(key):
   3766         key = list(key)
-> 3767     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   3769 # take() does not accept boolean indexers
   3770 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
 ↪5877, in Index._get_indexer_strict(self, key, axis_name)
   5874 else:
   5875     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 5877 self._raise_if_missing(keyarr, indexer, axis_name)
   5879 keyarr = self.take(indexer)
   5880 if isinstance(key, Index):
   5881     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
 ↪5938, in Index._raise_if_missing(self, key, indexer, axis_name)
   5936     if use_interval_msg:
   5937         key = list(key)
-> 5938     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   5940 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
   5941 raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['location'], dtype='object')] are in the [columns]"
```

[37]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[37], line 2
      1 encoder = OrdinalEncoder()
----> 2 df[categorical_column] = encoder.fit_transform(df[[categorical_column]]
```

```
File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:3767, in␣
↪DataFrame.__getitem__(self, key)
   3765     if is_iterator(key):
   3766         key = list(key)
-> 3767     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   3769 # take() does not accept boolean indexers
   3770 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
↪5877, in Index._get_indexer_strict(self, key, axis_name)
   5874 else:
   5875     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 5877 self._raise_if_missing(keyarr, indexer, axis_name)
   5879 keyarr = self.take(indexer)
   5880 if isinstance(key, Index):
   5881     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:
↪5938, in Index._raise_if_missing(self, key, indexer, axis_name)
   5936     if use_interval_msg:
   5937         key = list(key)
-> 5938     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   5940 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
   5941 raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['location'], dtype='object')] are in the [columns]"
```

[38]:
```
print(df.columns)
```

```
Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
       'price'],
      dtype='object')
```

[39]:
```
df.columns = df.columns.str.strip()  # Remove any leading/trailing spaces
df.columns = df.columns.str.lower()  # Convert all column names to lowercase␣
↪(optional)
print(df.columns)
```

```
Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
       'price'],
      dtype='object')
```

[40]:
```
# Optional: Use a conditional check
if 'location' in df.columns:
    print(df['location'].head())
else:
    print("The 'location' column is not in the DataFrame.")
```

```
0    a
1    b
2    c
3    d
4    e
Name: location, dtype: object
```

[41]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

[42]:
```
df
```

[42]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	dhaka	3	2	x	o+	10000
1	1.0	dhaka	4	2	y	ab+	12000
2	2.0	khulna	2	2	p	ab-	11000
3	3.0	rajshahi	3	2	q	o+	15000
4	4.0	cumilla	3	1	w	ab+	12000
5	5.0	dhaka	2	1	l	ab-	11000
6	6.0	barishal	2	1	r	o+	10000
7	7.0	natore	2	3	s	o+	15000
8	8.0	khulna	3	3	k	ab+	12000
9	9.0	dhaka	4	2	v	ab-	11000

4 Encoding / Handling another column => area

[43]:
```
categorical_column = 'area'
```

[44]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

[45]:
```
df
```

[45]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	x	o+	10000
1	1.0	2.0	4	2	y	ab+	12000
2	2.0	3.0	2	2	p	ab-	11000
3	3.0	5.0	3	2	q	o+	15000
4	4.0	1.0	3	1	w	ab+	12000
5	5.0	2.0	2	1	l	ab-	11000
6	6.0	0.0	2	1	r	o+	10000
7	7.0	4.0	2	3	s	o+	15000
8	8.0	3.0	3	3	k	ab+	12000
9	9.0	2.0	4	2	v	ab-	11000

5 Encoding more than one column at a time

[46]:
```
categorical_columns = ['owner', 'blodd group']  # Add or replace with your␣
↪desired column names
```

[47]:
```
for column in categorical_columns:
    if column not in df.columns:
        print(f"Column '{column}' does not exist in the DataFrame")
    else:
        print(f"Encoding column: {column}")
```

```
Encoding column: owner
Encoding column: blodd group
```

[48]:
```
encoder = OrdinalEncoder()
existing_categorical_columns = [col for col in categorical_columns if col in df.
↪columns]
df[existing_categorical_columns] = encoder.
↪fit_transform(df[existing_categorical_columns])
```

[49]:
```
df
```

[49]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	8.0	2.0	10000
1	1.0	2.0	4	2	9.0	0.0	12000
2	2.0	3.0	2	2	2.0	1.0	11000
3	3.0	5.0	3	2	3.0	2.0	15000
4	4.0	1.0	3	1	7.0	0.0	12000
5	5.0	2.0	2	1	1.0	1.0	11000
6	6.0	0.0	2	1	4.0	2.0	10000
7	7.0	4.0	2	3	5.0	2.0	15000
8	8.0	3.0	3	3	0.0	0.0	12000
9	9.0	2.0	4	2	6.0	1.0	11000

[50]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

[51]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
 price          1.000000
area           0.797921
bathroom       0.495798
blodd group    0.148712
location       0.133118
```

```
bedroom       -0.015721
owner         -0.092159
Name: price, dtype: float64
```

In [28]:
```
import pandas as pd

# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

In [29]:
```
df
```

Out[29]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

In [30]:
```
# Check for non-numeric data in the columns
print(df.dtypes)
```

```
location       object
area           object
bedroom         int64
bathroom        int64
owner          object
blodd group    object
price           int64
dtype: object
```

In [31]:
```
# Identify non-numeric values in specific columns
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

```
Non-numeric values in 'location ':
 ['a' 'b' 'c' 'd' 'e' 'f' 'g' 'h' 'i' 'j']
Non-numeric values in 'area ':
 ['dhaka' 'khulna' 'rajshahi' 'cumilla' 'barishal' 'natore']
Non-numeric values in 'owner ':
 ['x' 'y' 'p' 'q' 'w' 'l' 'r' 's' 'k' 'v']
Non-numeric values in 'blodd group':
 ['o+' 'ab+' 'ab-']
```

Encoding / Handling another column => location#

In [32]:
```
from sklearn.preprocessing import OrdinalEncoder
```

In [36]:
```
categorical_column = 'location'
```

In [35]:
```
df
```

Out[35]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

In [34]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[34], line 2
      1 encoder = OrdinalEncoder()
----> 2 df[categorical_column] = encoder.fit_transform(df[[categorical_colum
n]])

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:3767, in
DataFrame.__getitem__(self, key)
   3765     if is_iterator(key):
   3766         key = list(key)
-> 3767     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   3769 # take() does not accept boolean indexers
   3770 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:5
877, in Index._get_indexer_strict(self, key, axis_name)
   5874 else:
   5875     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 5877 self._raise_if_missing(keyarr, indexer, axis_name)
   5879 keyarr = self.take(indexer)
   5880 if isinstance(key, Index):
   5881     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:5
938, in Index._raise_if_missing(self, key, indexer, axis_name)
   5936     if use_interval_msg:
   5937         key = list(key)
-> 5938     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   5940 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique
())
   5941 raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['location'], dtype='object')] are in the [column
s]"
```

In [37]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[37], line 2
      1 encoder = OrdinalEncoder()
----> 2 df[categorical_column] = encoder.fit_transform(df[[categorical_colum
n]])

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:3767, in
DataFrame.__getitem__(self, key)
   3765     if is_iterator(key):
   3766         key = list(key)
-> 3767     indexer = self.columns._get_indexer_strict(key, "columns")[1]
   3769 # take() does not accept boolean indexers
   3770 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:5
877, in Index._get_indexer_strict(self, key, axis_name)
   5874 else:
   5875     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 5877 self._raise_if_missing(keyarr, indexer, axis_name)
   5879 keyarr = self.take(indexer)
   5880 if isinstance(key, Index):
   5881     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:5
938, in Index._raise_if_missing(self, key, indexer, axis_name)
   5936     if use_interval_msg:
   5937         key = list(key)
-> 5938     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
   5940 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique
())
   5941 raise KeyError(f"{not_found} not in index")

KeyError: "None of [Index(['location'], dtype='object')] are in the [column
s]"
```

In [38]:
```
print(df.columns)
```

```
Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
       'price'],
      dtype='object')
```

In [39]:
```
df.columns = df.columns.str.strip()  # Remove any leading/trailing spaces
df.columns = df.columns.str.lower()  # Convert all column names to lowercase (
print(df.columns)
```

```
Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
       'price'],
      dtype='object')
```

In [40]:
```
# Optional: Use a conditional check
if 'location' in df.columns:
    print(df['location'].head())
else:
    print("The 'location' column is not in the DataFrame.")
```

```
0    a
1    b
2    c
3    d
4    e
Name: location, dtype: object
```

In [41]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

In [42]:
```
df
```

Out[42]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	dhaka	3	2	x	o+	10000
1	1.0	dhaka	4	2	y	ab+	12000
2	2.0	khulna	2	2	p	ab-	11000
3	3.0	rajshahi	3	2	q	o+	15000
4	4.0	cumilla	3	1	w	ab+	12000
5	5.0	dhaka	2	1	l	ab-	11000
6	6.0	barishal	2	1	r	o+	10000
7	7.0	natore	2	3	s	o+	15000
8	8.0	khulna	3	3	k	ab+	12000
9	9.0	dhaka	4	2	v	ab-	11000

Encoding / Handling another column => area

In [43]:
```
categorical_column = 'area'
```

In [44]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

In [45]:
```
df
```

Out[45]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	x	o+	10000
1	1.0	2.0	4	2	y	ab+	12000
2	2.0	3.0	2	2	p	ab-	11000
3	3.0	5.0	3	2	q	o+	15000
4	4.0	1.0	3	1	w	ab+	12000
5	5.0	2.0	2	1	l	ab-	11000
6	6.0	0.0	2	1	r	o+	10000
7	7.0	4.0	2	3	s	o+	15000
8	8.0	3.0	3	3	k	ab+	12000
9	9.0	2.0	4	2	v	ab-	11000

Encoding more than one column at a time

In [46]:
```
categorical_columns = ['owner', 'blodd group']  # Add or replace with your des
```

In [47]:
```
for column in categorical_columns:
    if column not in df.columns:
        print(f"Column '{column}' does not exist in the DataFrame")
    else:
        print(f"Encoding column: {column}")
```

```
Encoding column: owner
Encoding column: blodd group
```

In [48]:
```
encoder = OrdinalEncoder()
existing_categorical_columns = [col for col in categorical_columns if col in d
df[existing_categorical_columns] = encoder.fit_transform(df[existing_categoric
```

In [49]:
```
df
```

Out[49]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	8.0	2.0	10000
1	1.0	2.0	4	2	9.0	0.0	12000
2	2.0	3.0	2	2	2.0	1.0	11000
3	3.0	5.0	3	2	3.0	2.0	15000
4	4.0	1.0	3	1	7.0	0.0	12000
5	5.0	2.0	2	1	1.0	1.0	11000
6	6.0	0.0	2	1	4.0	2.0	10000
7	7.0	4.0	2	3	5.0	2.0	15000
8	8.0	3.0	3	3	0.0	0.0	12000
9	9.0	2.0	4	2	6.0	1.0	11000

In [50]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

In [51]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
 price          1.000000
area           0.797921
bathroom       0.495798
blodd group    0.148712
location       0.133118
bedroom       -0.015721
owner         -0.092159
Name: price, dtype: float64
```

In [ ]: